

Workshop 1 (W1): Discovering Questions Worth Investigating

Comparative Report and Selection of a Research Topic for the Chilean Health Observatory

Course: Business Intelligence (IIB423T-1) — Universidad del Desarrollo

Instructor: Tomás Fontecilla Correa

Team members: Benjamín P. and Sebastián Herrera

Submission date: September 25, 2026

Research topics assessed: Candidate A (selected), seasonality of respiratory emergency visits and winter service capacity (2020–2024); Candidate B (alternative), nutritional status of young children receiving care in the public health network (REM-P2, 2020–2024).

1. Executive Summary and Methodological Scope

This report documents the analysis and business framing used to select a research topic for the main course project (C1). To contribute to an educational observatory of Chilean health data, the team examined two public health questions using official datasets from the Department of Health Statistics and Information (DEIS) and the Ministry of Health (MINSAL):

1. **Candidate A: The weekly demand for emergency care due to respiratory causes across 52 or 53 weeks per year from 2020 to 2024, with a focus on Winter Campaign staffing and bed capacity.**
 2. **Candidate B: Trends in overweight and obesity among children under six receiving routine care in the public network, using the June and December REM-P2 reporting periods.**
-

Methodological caution: Both analyses use administrative counts of visits and health checks reported by public providers. They do not estimate population prevalence, municipal incidence, individual transitions, or clinical causation. Their measures describe activity recorded in the health network and depend on facility reporting.

After a Five Cs data quality audit, sensitivity analysis, and diagnostic reconciliation of more than 42 million emergency records and 4.6 million REM records, the team provisionally selected Candidate A. It offers a continuous, high frequency series covering 262 weeks, complete weekly coverage across five years, a direct link to Winter Campaign decisions, and a sound basis for future forecasting and operational dashboards.

2. Candidate A (Selected): Respiratory Emergency Visits and Winter Capacity

2.1. The Problem and Its Scope

- **Service concern:** Seasonal pressure on hospital and primary care emergency services in Chile as respiratory viruses circulate, including respiratory syncytial virus (RSV), influenza A/B, adenovirus, metapneumovirus, and SARS-CoV-2.
 - **Population and service:** People attending the public emergency care network, including hospitals, SAR, SAPU, and SUR facilities.
 - **Geographic and time coverage:** National data by facility and epidemiological reporting week (weeks 1–52/53) for the full 2020–2024 period.
-

2.2. Seven Step Business Framing

1. **Decision to support:** Develop an analytical monitoring tool for the Directorate for Health Care Network Management (DIGERA), Health Service directors, and emergency department leaders planning the Winter Campaign. It could guide the timing of additional physician, nursing, and respiratory therapy shifts; the phased conversion of general beds to pediatric and adult critical care beds; and coordination of primary care emergency capacity (SAPU/SAR) before hospital emergency departments become overloaded.
 2. **Success criteria:** Build a consistent 2020–2024 series; identify the onset, peak, and decline of winter demand; calculate the respiratory share of all emergency visits; describe the age profile of infants, children aged 1–4, and adults aged 65 or more; audit the Five Cs; and confirm that diagnostic categories remain stable.
 3. **Business question:** How did the weekly volume and age composition of respiratory emergency visits in Chile vary from 2020 to 2024 by facility type and region? How did seasonality and magnitude differ between the pandemic years (2020–2021) and the later years (2022–2024)?
 4. **Care process:** Each provider reports daily Emergency Care Records (DAU) through the Emergency Care System (SADU) to DEIS.
 5. **Entity, event, and state:** The entity is a health facility; the event is a recorded emergency visit on a given date for a diagnostic cause; and the state is an aggregate daily or weekly count. The dataset has no patient identifiers, so repeat visits by one person cannot be identified.
 6. **Representation:** One row per facility, date, week, and cause code (IdCausa). Measures are Total and age bands Menores_1, De_1_a_4, De_5_a_14, De_15_a_64, and De_65_y_mas. GLOSATIPOESTABLECIMIENTO and GLOSATIPOATENCION classify the type of facility and care.
 7. **Measures:**
 8. *Primary measure 1: Weekly respiratory emergency visits (IdCausa = 2).*
 9. *Primary measure 2: Respiratory share = $\text{Total}(\text{IdCausa} = 2) / \text{Total}(\text{IdCausa} = 1)$, the percentage of all emergency visits attributed to respiratory causes.*
 10. *Secondary measures: Percentage distributions by age band and care level (hospitals versus SAPU/SAR/SUR).*
-

2.3. Official Documentary Support (MINSAL / DEIS)

- **MINSAL (2024), Winter Campaign 2024: Strategy to Strengthen the Health Care Network and Epidemiological Monitoring.** The Ministry’s Health Care Networks Secretariat describes hospital resource planning and weekly emergency monitoring to reduce shortages and critical bed pressure. (2024 campaign link in references.)
- **DEIS–MINSAL (2024), Weekly Epidemiological Surveillance of Respiratory Viruses and Emergency Visits.** DEIS publishes the weekly monitoring series used to assess emergency demand. (DEIS link in references.)
- **MINSAL (2022), National Health Strategy for Health Objectives to 2030,** including communicable diseases and acute respiratory infections. (Strategy link in references.)

2.4. Inspection of Actual Data and Five Cs Audit

The official compressed files `AtencionesUrgenciaYYYY.zip` (2020–2024) were scanned in chunks of 250,000 rows and aggregated as they were read. In total, 42,061,502 raw rows were inspected. The target cause codes account for 12,967,058 rows: codes 1 and 2, respiratory subcauses 3, 4, 5, 6, 10, and 11, and COVID-19 codes 30–33. The year by year inventory follows.

Year	Source ZIP	Raw rows	Selected cause rows	Cause coverage (%)
2020	AtencionesUrgencia2020.zip	6,446,646	2,282,606	35.41%
2021	AtencionesUrgencia2021.zip	8,816,240	2,644,872	30.00%
2022	AtencionesUrgencia2022.zip	8,926,307	2,677,892	30.00%
2023	AtencionesUrgencia2023.zip	8,899,080	2,669,724	30.00%
2024	AtencionesUrgencia2024.zip	8,973,229	2,691,963	30.00%

Formal Data Quality Audit Using the Five Cs

1. **Completeness:** All epidemiological reporting weeks are present: weeks 1–53 in 2020 and 2021, and weeks 1–52 in 2022–2024. Total and all five age band columns have no missing values across the five years.
2. **Consistency:** For each of the 12.9 million selected rows, Total was compared with the sum of the five age bands. No arithmetic mismatches were found.
3. **Conformity and validity:** No negative values were found in the numeric columns and all weeks fall within [1, 53]. Cause identifiers match Annex 1 of the SADU data dictionary.
4. **Uniqueness:** A full key audit of the selected rows found no duplicate facility/date/cause keys in 2020–2022, 12 extra key rows in 2023, and three in 2024. The duplicates add 13 to the all-visit count in 2023 and zero to respiratory visits. The reported annual totals retain the raw source rows; this small anomaly should be resolved before facility-level C1 analysis.

- Comparability and currency:** The diagnostic codes and relevant fields are stable across the five years. The pandemic years 2020–2021 are analytically distinct from 2022–2024 because public health measures changed the usual winter pattern.

=== FIVE CS AUDIT SUMMARY: RESPIRATORY EMERGENCY VISITS ===

Year weeks	Rows inspected Duplicate keys	Weeks	Missing Total	Age sum mismatches	Negatives	Invalid
2020 0	2,282,606	1–53 (53)	0	0	0	0
2021 0	2,644,872	1–53 (53)	0	0	0	0
2022 0	2,677,892	1–52 (52)	0	0	0	0
2023 12	2,669,724	1–52 (52)	0	0	0	0
2024 3	2,691,963	1–52 (52)	0	0	0	0

2.5. Empirical Findings

A. Annual Totals and Changes in Respiratory Demand

Year	All emergency visits (ld 1)	Respiratory visits (ld 2)	Respiratory share	Epidemiological context
2020	11,382,496	1,223,447	10.75%	Lockdowns, remote work, and masks suppressed usual seasonal virus circulation.
2021	13,851,594	1,381,568	9.97%	Containment measures continued; circulation of traditional viruses remained low.
2022	19,491,232	4,812,171	24.69%	Schools returned to in person attendance and acute respiratory infections rebounded.
2023	19,194,240	5,557,541	28.95%	An early, severe RSV outbreak affected infants.
2024	19,436,048	5,541,628	28.51%	High winter demand persisted; respiratory causes represented about 28.5% of emergency activity.

B. Peak Winter Weeks

- 2020:** An unusual early peak in week 12 (March), with 103,314 respiratory visits before lockdown; winter demand then remained relatively flat.
- 2021:** A late peak in week 46 (November), with 47,697 visits.

- 2022: A peak in week 44 (November), with 161,522 visits following renewed viral circulation in spring.
- 2023: A sharp winter peak in week 20 (May), with 189,297 respiratory visits in one week, accounting for 41.66% of all emergency visits that week.
- 2024: The winter peak occurred in week 20 (May), with 203,353 respiratory visits and a 45.56% respiratory share. Operational implication: In the postpandemic period, weeks 20–32 (roughly May–August) form the critical window for seasonal capacity planning.

C. Age Distribution: Pediatric and Older Adult Demand

- Under one year: 5.06% (936,038 visits). This group faces particular risk from bronchiolitis and may require pediatric critical care.
- Ages 1–4: 16.16% (2,992,646 visits).
- Ages 5–14: 23.11% (4,278,843 visits).
- Ages 15–64: 46.67% (8,640,927 visits).
- Ages 65 and over: 9.01% (1,667,901 visits). Together, children under 15 account for 44.33% of respiratory emergency visits.

D. Care Level: Where Winter Demand Is Handled

Year	Hospital	SAPU	SAR	SUR
2020	28.46%		48.09%	19.36%
2022	27.73%		41.94%	23.12%
2024	24.13%		42.28%	25.76%

Management finding: Primary emergency care (SAPU + SAR + SUR) handles 75.87% of national respiratory emergency demand. Hospital pressure may rise when this network cannot absorb outpatient demand during the winter peak. This aggregate share does not by itself establish a causal pathway.

2.6. Sensitivity Analysis and Cause Reconciliation

1. COVID-19 sensitivity: The official respiratory series (IdCausa 2) was compared with an expanded series adding COVID-19 codes 30–33. In 2024, the respiratory share rises from 28.51% to 28.79%, a difference of 0.28 percentage points. In 2021, adding COVID-19 raises it from 9.97% to 25.50%, a difference of 15.53 points. This supports reporting COVID-19 separately when comparing years.
2. Subcause reconciliation: The sum of bronchitis (3), influenza (4), pneumonia (5), other respiratory causes (6), upper acute respiratory infection (10), and obstructive episodes (11) was compared with IdCausa 2. The national totals

match exactly in every year from 2020 to 2024, supporting consistency of this aggregate classification.

3. Proposed join (PROPOSED): To add municipality and region to 2020–2022, when these fields are not provided directly as they are in 2023–2024, join IdEstablecimiento to the relevant annual establishment register Establecimientos/Base_Establecimientos_YYYY.xlsx. This join has not been executed in the reported results.
-
-
-

3. Candidate B (Alternative): Child Nutritional Status in REM-P2

3.1. The Problem and Its Scope

- Service concern: Excess weight, including overweight and obesity, among Chilean children receiving routine health checks in primary care.
 - Population and service: Children under six enrolled in and assessed through the public child health program.
 - Geographic and time coverage: National data reported for June and December in each year from 2020 to 2024.
-

3.2. Seven Step Business Framing

1. Decision to support: Provide child health teams with a monitoring view to prioritize data quality audits and focus nutrition promotion in primary care facilities.
 2. Success criteria: Confirm service codes and age bands in the annual Series P dictionaries; quantify coverage of matched facilities; show comparable reporting periods with quality caveats; and chart the time series.
 3. Business question: Among children recorded in REM-P2, how did the combined count for overweight (P2070503) and obesity (P2070504) compare with the population under care denominator (P2060000, Col01, both sexes) in the June and December reporting periods of 2020–2024?
 4. Care process: Routine child health checks, including anthropometric assessment, are consolidated as aggregate counts in the P2 sheet of REM Series P.
 5. Entity, event, and state: The entity is a health facility, the event is a recorded child health check, and the state is an aggregate count for a six month reporting point. The data do not follow individuals; a child may appear in
-

both periods of a year. These are proportions of recorded children under care, not population prevalence.

6. Representation: One row by year, reporting month (6 or 12), health service, facility, and service code. Col01 records both sexes combined; Col04–Col31 break counts down by age and sex.
7. Measure: $(P2070503 \text{ Col01} + P2070504 \text{ Col01}) / P2060000 \text{ Col01}$, calculated only for facility-period units reporting the denominator and both numerator codes.

3.3. Documentary Support

- MINSAL (2023), Strategy to Halt the Acceleration of Overweight and Obesity in Children and Adolescents 2023–2030, sets nutrition policy objectives. Those objectives should not be interpreted as observed outcomes in administrative records. (Strategy link in references.)

3.4. Inspection of Actual Data and Key Limitations

The annual SERIE_REM_YYYY.zip files range from 455,729 rows in 2020 to 1,200,514 rows in 2024. Inspection found four major limitations. (1) June 2020 coverage collapse: Only 83 facilities reported a P2060000 denominator, representing 51,788 children, compared with more than 1,950 facilities and 550,000 children in other periods. June 2020 is not nationally comparable. (2) Missing age and sex breakdowns: In 2024, only 713 of 1,764 matched units (40.4%) had all 26 age and sex fields populated, so the main analysis uses Col01. (3) Changing codes: P2501800, for children without assessment owing to a special condition, appears only in the 2023 and 2024 dictionaries; category reconciliation across all five years is therefore limited. (4) Duplicate keys: In December 2021, 18 duplicate rows at two facilities required analytical deduplication.

=== CANDIDATE B: OVERWEIGHT + OBESITY SHARE AMONG CHILDREN UNDER CARE ===

Year	June	December	Data quality note
2020	33.61%*	35.05%	*June severely underreported: only 83 facilities.
2021	35.46%	37.53%	Duplicates detected in December.
2022	36.59%	35.46%	Aggregate Col01 counts relatively stable.
2023	34.59%	34.76%	Code P2501800 first appears.
2024	35.16%	34.72%	Complete age breakdown in only about 40% of matched units.

4. Critical Comparison: Candidate A versus Candidate B

The following matrix compares the topics on the five dimensions required by the W1 rubric.

Comparison dimension	Candidate A: Respiratory emergency visits (selected)	Candidate B: REM-P2 child nutrition (alternative)
1. Relevance supported by official sources	Very high: Supported by the 2024 MINSAL Winter Campaign, DEIS weekly reports, and the 2030 National Health Strategy. Directly relevant to emergency staffing and bed planning.	High: Supported by MINSAL’s 2023 –2030 childhood obesity strategy. Policy targets, however, cannot be treated as outcomes observed in administrative records.
2. Fit between question and inspected data	Excellent: Continuous daily and weekly granularity, with 52/53 weeks per year, supports seasonal curves, peaks, age bands, and care level comparisons.	Limited: Only two reporting points per year (ten in five years). June 2020 has severely reduced coverage and is not nationally comparable.
3. Analytical opportunity for C1	Strong: Supports time series, epidemiological curves, care level comparisons, and pandemic versus later period analysis over millions of selected rows.	Moderate: Largely limited to descriptive comparisons across six monthly Col01 snapshots; monthly trends cannot be reconstructed.
4. Main limitation or uncertainty	Pandemic measures in 2020–2021 changed usual winter viral circulation, complicating historical baselines.	Severe June 2020 coverage gap, more than 59% missing age/sex detail, and the introduction of P2501800 only in 2023.
5. Feasible scope and future ML/dashboard use	High: Weekly data could support a two to four week forecast and an operational dashboard for emergency pressure, subject to validation and added predictors.	Low for ML: Ten six monthly points are insufficient for a robust predictive model; a descriptive annual dashboard is feasible.

5. Rationale for the Provisional Choice and Questions for the Instructor

5.1. Why Candidate A Was Chosen

The team provisionally favors Candidate A for three reasons. (1) Temporal density and continuity: Emergency data provide 262 continuous weekly observations; REM-P2 provides ten six monthly reporting points. The weekly series supports a richer C1 exploration and possible later forecasting. (2) Data completeness: The emergency series has no missing values in the audited fields and complete weekly coverage,

while REM-P2 has a major June 2020 coverage failure and missing values in more than 59% of age and sex fields in the reported comparison. (3) Operational relevance: Primary emergency care handles 75.87% of respiratory visits, making its winter capacity directly relevant to planning.

What could change the choice? The team would reconsider REM-P2 if the instructor or DEIS could identify a validated monthly series that resolves the June 2020 gap and an official rule for handling missing age and sex cells. On the currently available evidence, respiratory emergency visits provide the stronger research basis.

5.2. Specific Questions for Instructor Feedback

1. **Pandemic shock in the baseline:** Given that lockdowns and masks changed usual viral circulation in 2020–2021, should a historical seasonal baseline exclude 2020 or include an explicit pandemic indicator?
 2. **Geographic approach for C1:** Emergency data contain municipality and region directly in 2023–2024, while 2020–2022 require a join to the establishment register. Should C1 prioritize a national five year analysis or a more detailed geographic analysis limited to 2023–2024?
 3. **Primary care capacity:** Given that primary emergency facilities handle more than three quarters of respiratory visits, would focusing the C1 business question on their capacity to absorb demand be appropriate?
-

6. Project Continuation Beyond C1: Machine Learning and Dashboard

6.1. Possible Machine Learning Extension

- **Candidate A: Respiratory Emergency Visits**
 - *Proposed task:* Forecast weekly respiratory emergency volume two to four weeks ahead by Health Service or facility type using a multivariable time series model.
 - *Potential use:* Give earlier warning for pediatric critical bed conversion and staffing before emergency demand peaks.
 - *Additional data needed:* Weekly positivity for RSV and influenza from the Public Health Institute (ISP), and open weekly minimum temperature data.
-

- *Evaluation: Compare with a lag one and seasonal moving average baseline using MAE and RMSE on a time ordered split (training 2020–2023; test 2024).*
 - **Candidate B: REM-P2**
 - *Machine learning limitation: Ten six monthly aggregate points are too few to train and validate a robust forecasting or classification model without severe overfitting. A claim to apply machine learning here would be premature.*
-

6.2. Business Intelligence Dashboard Extension

- **Candidate A: Respiratory Emergency Visits**
 - *Users: Emergency department leaders, Health Service directors, and DIGERA bed management teams.*
 - *Decisions: Monitor the weekly demand curve against care capacity and activate contingency measures in stages.*
 - *Views and indicators: (1) Observed weekly curve against a five year historical band; (2) respiratory share alert indicator, with a proposed threshold above 30%; (3) age distribution for infants under one and adults aged 65 or more; (4) share of demand handled by SAPU/SAR versus hospitals. The alert threshold is a proposed design choice, not an official standard.*
 - **Candidate B: REM-P2**
 - *Descriptive dashboard: An annual view for family health centre (CESFAM) directors comparing the overweight/obesity share across facilities using Col01, with explicit warnings for underreporting in individual periods.*
-
-
-

7. Process Record

This section provides the decision trail, contributions, and AI use required by the W1 rubric.

7.1. Decisions and Methodological Checks

Date / stage	Consequential decision	Alternative considered	Reason and supporting evidence	Check performed	File location
Sep 24 / source retrieval	Retrieve official DEIS sources programmatically.	Download isolated files manually.	Preserve reproducibility, SHA-256 hashes, and version provenance.	Check ZIP integrity and record downloads in CSV.	data/source_manifest.csv
Sep 25 / Candidate A	Aggregate the source files in streaming chunks.	Load all raw DataFrames into memory.	Loading 42 million raw rows exhausted available memory; aggregation while reading greatly reduced memory use.	Compare memory use and verify exact integer totals.	analysis/urgencias_respiratorias.ipynb (cells 3–5)
Sep 25 / Candidate A	Keep COVID-19 codes 30–33 separate from IdCausa 2.	Add COVID-19 directly to the official respiratory total.	COVID-19 adds 0.28 percentage points in 2024 and is classified separately by DEIS.	Compare series with and without COVID-19.	analysis/urgencias_respiratorias.ipynb (cells 10–11)
Sep 25 / Candidate B	Use Col01 as the main REM-P2 measure.	Use the age and sex breakdown in Col04–Col31.	More than 59% of matched units lack complete age/sex detail; forcing those fields would distort comparison.	Count missing P2 cells.	analysis/rem_p2.ipynb (cells 12–14)
Sep 25 / selection	Choose Candidate A for C1.	Continue both topics or choose REM-P2.	Emergency data provide 262 weekly observations versus ten REM reporting points and can support a winter capacity dashboard.	Review the five dimension comparison matrix.	report/GROUP_XX_W1_Report.pdf (§4)

7.2. Instructor Feedback and Response

- **Status:** Pending the guided workshop. The three questions in Section 5.2 are intended for discussion with Professor Tomás Fontecilla.
-

7.3. Individual Contributions

- **Benjamín P.:** Led the analytical framing of Candidate A; designed chunk based streaming to process approximately 42 million emergency records without exhausting memory; performed the Five Cs audit and COVID-19 sensitivity analysis; made the multiyear visualizations; and drafted the integrated comparison.
 - **Sebastián Herrera:** Led the Candidate B analysis; inspected Series P codebooks with openpyxl; identified the June 2020 coverage collapse and the introduction of P2501800; set up the repository source inventory; and drafted the child nutrition discussion.
-

7.4. AI Use and Verification

- **Tools used:** AI assistants (Gemini / Antigravity and Codex).
 - **Purpose:** Help with chunked reading of compressed files, Matplotlib chart syntax, the decision matrix text, and table formatting.
 - **Examples and verification:** An initial suggestion to load all raw DataFrames was rejected in favor of streaming aggregation. For this delivery, the Codex request was to format tables, download a copy, organize files to the brief, and translate the materials into English. Formatting and structure changes were retained after checking numerical differences against executed notebooks and verifiable extracts. The exported PDF was visually inspected and the ZIP checked for a single root folder.
-

7.5. Individual Reflections

- **Benjamín P. – reflection:**
 - *Contribution:* Efficient analysis of more than 42 million emergency records and the Five Cs quality audit.
 - *Learning:* Administrative care counts differ from population prevalence; primary emergency facilities (SAPU/SAR/SUR) handle more than 75% of recorded respiratory visits.
 - *Remaining uncertainty:* How best to account for the 2020–2021 pandemic shock in future time series models.
 - **Sebastián Herrera – reflection:**
 - *Contribution:* Audit of Series P codebooks and identification of reporting coverage bias in REM-P2.
-

- *Learning: A public policy objective does not ensure that administrative data measure it without bias.*
 - *Remaining uncertainty: Confirming with DEIS what absent code rows mean in the monthly summaries.*
-
-
-

8. Official References

1. DEIS–MINSAL (2024). Open Data: Emergency Care Records from the Health Information System (SADU), 2020–2024. Department of Health Statistics and Information, Chilean Ministry of Health. Available at: <https://deis.minsal.cl/#datosabiertos>
 2. DEIS–MINSAL (2024). Data Dictionary: Emergency Care Records in the Emergency Care System. Department of Health Statistics and Information, Chilean Ministry of Health.
 3. DEIS–MINSAL (2024). Monthly Statistical Summaries (REM Series P), 2020–2024. Department of Health Statistics and Information, Chilean Ministry of Health.
 4. MINSAL (2024). Winter Campaign 2024: Strategy to Strengthen the Health Care Network and Epidemiological Surveillance. Health Care Networks Secretariat, Chilean Ministry of Health. Available at: <https://www.minsal.cl/campana-de-invierno-2024/>
 5. MINSAL (2023). Strategy to Halt the Acceleration of Overweight and Obesity in Children and Adolescents, 2023–2030. Chilean Ministry of Health. Available at: <https://www.minsal.cl/estrategia-sobrepeso-obesidad-2023-2030/>
 6. MINSAL (2022). National Health Strategy for Health Objectives to 2030. Chilean Ministry of Health. Available at: <https://www.minsal.cl/estrategia-nacional-de-salud-2030/>
-